

SHRAVANKUMAR NAGARAJAN

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PROFESSIONAL SUMMARY

Versatile AI & Machine Learning Engineer experienced in designing data-driven software systems, predictive analytics, and intelligent automation. Skilled in Python, PyTorch, Scikit-learn, Spark, LangChain, and AWS SageMaker, with strong foundations in statistics, optimization, and deep learning. Proven ability to bridge ML modeling, scalable backend engineering, and MLOps to deliver measurable business outcomes across banking, healthcare, and analytics domains.

SKILLS

- **Programming Languages:** Python, SQL, JavaScript, Java, Bash, C/C++
- **Machine Learning:** Scikit-learn, Pytorch, XGBoost, Deep Learning, NLP, Computer Vision, Bayesian Inference
- **AI & GenAI:** LangChain, LangGraph, LlamaIndex, OpenAI APIs, Claude, Llama, Ollama, RAG Pipelines, Agentic AI
- **Big Data & Libraries:** PySpark, Airflow, Kafka, AWS Glue, Redshift, Databricks, Flink, Trino
- **MLOps & Cloud:** Docker, Kubernetes, Helm, AWS SageMaker, Concourse
- **Databases:** MySQL, Mongo, Cassandra, Elasticsearch, FAISS, ChromaDB, pgvector
- **Messaging & Streaming:** Apache Kafka, Amazon SQS, Amazon Kinesis
- **Visualization & Monitoring:** Grafana, Prometheus, ELK, Newrelic, AppDynamics
- **Mathematical Foundations:** Linear Algebra, Probability, Statistics, Optimization

PROJECTS & OUTSIDE EXPERIENCE

Healthcare Cost Analytics - MIMIC-IV Big Data Pipeline

- Built a Big Data analytics pipeline using the MIMIC-IV dataset to analyze how ICU costs vary with nurse-to-patient ratios, lab usage, and medication trends. Results showed a 22% cost increase when ratios dropped below 1:3, revealing the financial impact of understaffing and enabling predictive staffing optimization.
- Streamed simulated ICU events via Kafka 'Apache Flink, applying Bloom Filters for deduplication and HyperLogLog for scalable cardinality estimation, then transformed and warehoused data in Amazon Redshift via AWS Glue for OLAP dashboards and cost analytics.
- Leveraged XGBoost regression models to predict ICU costs based on real-time clinical and staffing variables, applying gradient-descent-based hyperparameter optimization to maximize model performance. Achieved $R^2 = 0.91$, MAE = 5.3 %, and cost-prediction precision = 93 %, validated via 5-fold cross-validation, demonstrating robust generalization and strong predictive reliability.

Job Analytics LinkedIn Job Insights

- Built a real-time LinkedIn job analytics platform that continuously scrapes and processes job postings to uncover skill, salary, and industry trends
- Implemented streaming ingestion with Python + Selenium, storing structured data in MySQL and MongoDB via Prisma ORM
- Developed a full-stack React + Vite UI for live dashboards and interactive analytics
- Integrated LangChain, LlamaIndex, and LangGraph agents for semantic search, natural-language querying, auto-summarization, and skill-demand insights over job data, achieving 92 % retrieval accuracy compared to keyword baselines.

PROFESSIONAL EXPERIENCE

The Bank of New York Mellon | *Lead Full Stack Developer*

Sep 2022 - Jan 2025

- Led design of high-throughput data pipelines (Kafka -> Spark -> ElasticSearch and Redshift) enabling real-time telemetry insights.
- Designed and deployed deep learning models (Autoencoder & LSTM) for detecting performance anomalies across application servers and databases and automated CI/CD deployment workflows for ML & analytics services using Concourse and Helm.
- Developed XGBoost-based predictive models for server capacity and utilization forecasting and anomaly detection, reducing false alerts by 35% and optimizing resource utilization.

Elsevier | *Senior Software Engineer*

Jul 2020 - Sep 2022

- Designed distributed semantic-search pipelines ingesting 120 TB+ documents via Kafka and PySpark into Elasticsearch, sustaining over 10 million document ingestions per day with less than 500 ms query latency across global search platforms
- Developed entity-recognition-driven contextual search APIs using SciBERT and spaCy NLP pipelines, which improved user relevance by 35% and reduced query latency by 40% through optimized vector indexing and model inference caching
- Developed AI-driven literature-recommendation modules combining keyword extraction, and transformer-based embeddings, enabling millions of researchers on Scopus and Mendeley to discover related work more efficiently.

EDUCATION

San Jose State University

Jan 2025 - Dec 2026

Master's, Applied Data Intelligence (GPA: 3.67)

- **Coursework:** Machine Learning, Deep Learning, Gen AI, Agentic AI